ATOM MARK 1



DESCRIPTION

The electric car designed and constructed by college students is an impressive blend of innovation, sustainability, and cutting-edge technology. Crafted with a combination of academic knowledge and hands-on expertise, this vehicle represents a remarkable achievement in the world of automotive engineering.

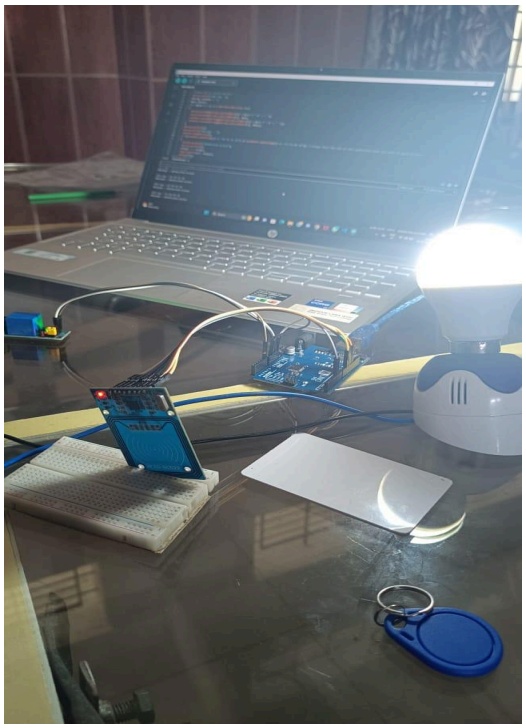
Components used:

- 2000 Watts BLDC Motor
- 60 V 24 AH Battery
- Motor Controller
- 42 inch Differential

Participants

1. Anuraj T R
2. Indhuchoodan R
3. Nikhil T Das

EDU BULB



DESCRIPTION

The LED Bulb is a remarkable achievement born from the creative minds and hard work of a group of visionary college students. This groundbreaking LED bulb is more than just a source of light; it's a testament to innovation, sustainability, and youthful ingenuity.

Components used:

- Punching Machine
- Tikki Fitting Machine
- Soldering Iron

Participants

1. Anuraj T R
2. Indhuchoodan R
3. Nikhil T Das

MGU Result Scrapper



```
const { chromium } = require("playwright");
const prnList = require('./data.json');

(async () => {
    let browser = await chromium.launch();
    let page = await browser.newPage();
    await page.setViewportSize({ width: 1000, height: 850 });

    await page.goto("https://parooksha.mgu.ac.in/Parooksha/index.php/Public/ParookshaResultView ctrl/index/3");
    // await page.goto("https://dodi.mgu.ac.in/ecOg4gat/index.php/public/ResultView_ctrl/");

    await page.selectOption('select#exam_id', prnList.exam_id);
    for (let i   0; i < prnList.prn.length; i++) {

        await page.fill('#prn', prnList.prn[i]);
        await page.click("button#btnresult");

        await page.waitForSelector('div#mgu_btech_contentholder table:nth-child(3)', { visible: true });
        await page.screenshot({
            path: `results/${prnList.prn[i]}.png`
        });
    }

    await browser.close();
})();
```

DESCRIPTION

It is the project to get the result screenshots of a list of students from the MGU result portal via email in the form of secure PDFs.

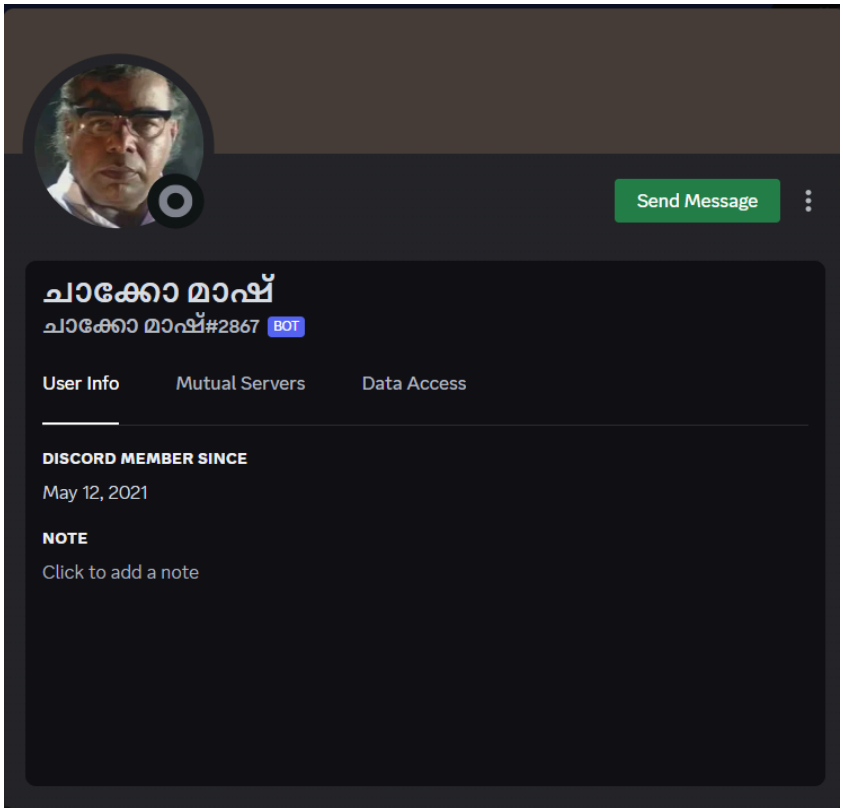
Components used:

- Playwright
- Nodemailer

Participants

1. Arjun Krishna

Chacko Mash



DESCRIPTION

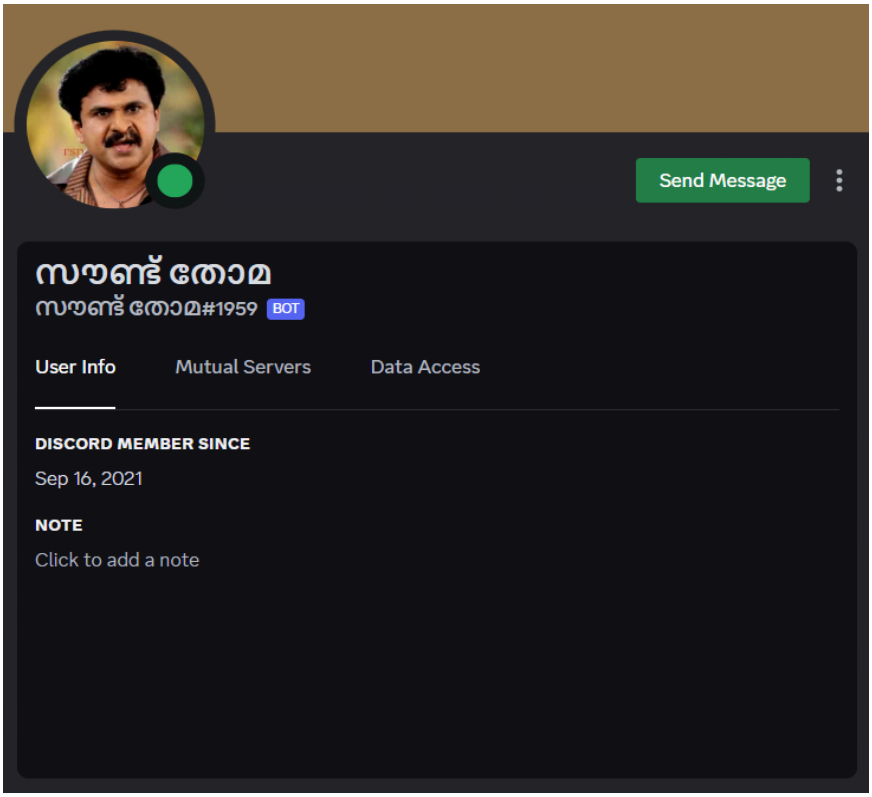
Chacko Mash is the official Discord Bot of Inovus Labs Discord Server. It's a custom-built Discord Bot to effectively and efficiently manage the Discord Server and to maintain it lively.

- Server used : Doku
- Language : Javascript

Participant

1. Arjun Krishna

Sound Thoma



DESCRIPTION

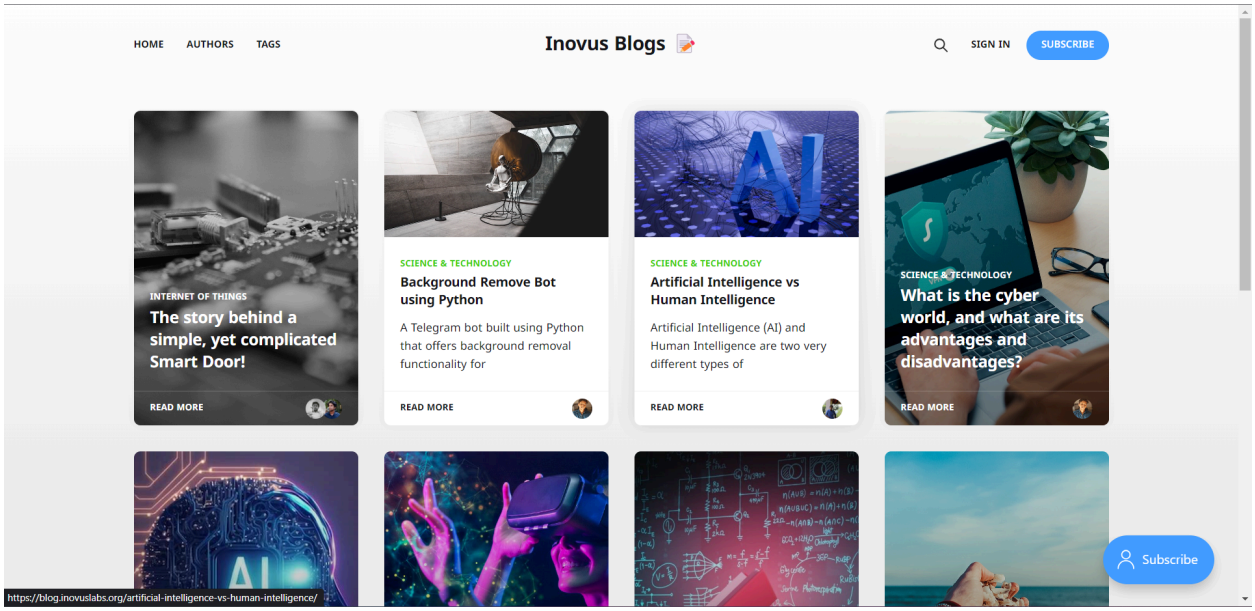
Sound Thoma is your go-to music bot for Discord, designed to make your server's music experience unforgettable.

- Server used : Doku
- Language : Javascript

Participant

1. Arjun Krishna

Inovus Blogs



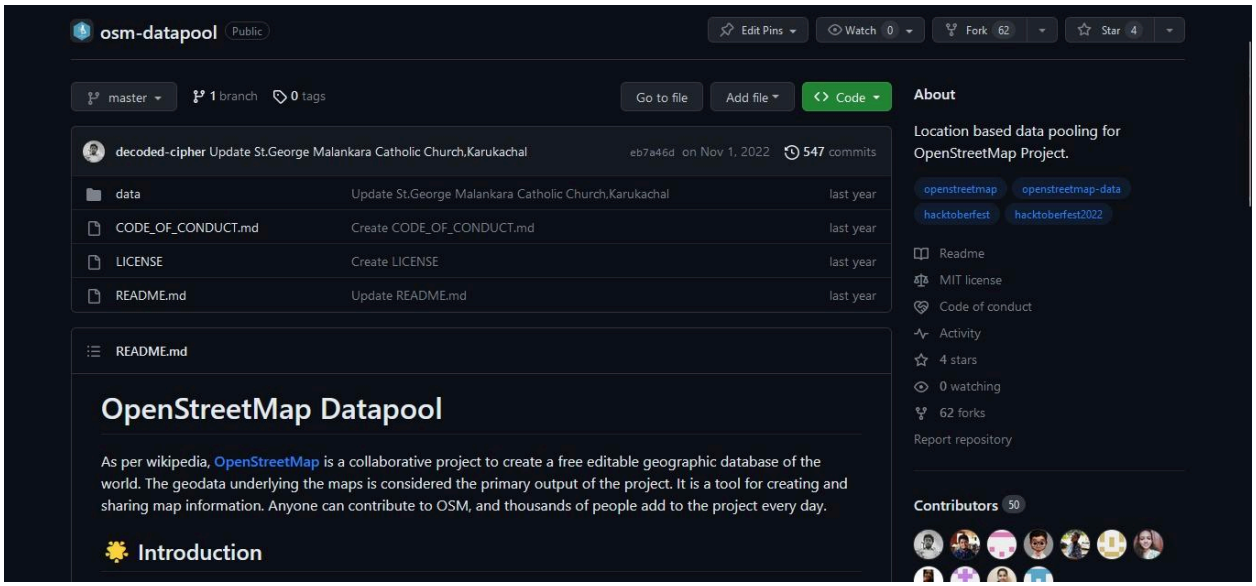
DESCRIPTION

Inovus Blogs are curated by students as a dynamic platform, where young minds converge to share their opinions,ideas,insights,research, and individual perspectives across a spectrum of disciplines. These blogs serve as a conduit for the academic prowess and creative thinking of students, fostering a culture of learning,art of expression,responsible citizenship and discipline of writing.

Participants

1. Arjun Krishna
2. Nikhil T Das

OSM Datapool



DESCRIPTION

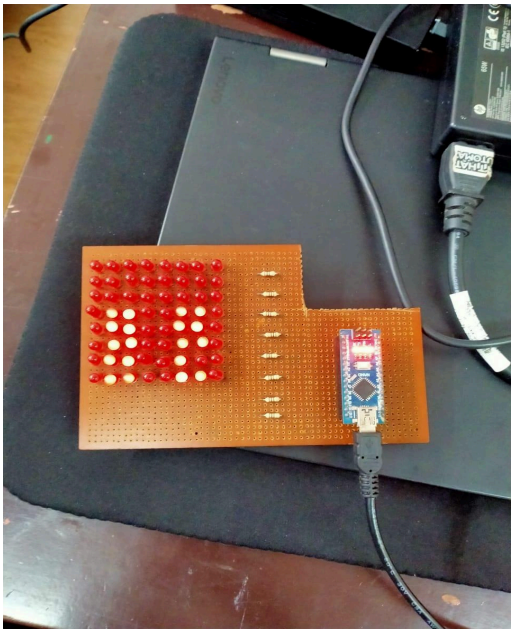
The OSM (OpenStreetMap) Data Pool is an ambitious open-source initiative undertaken by Inovus Labs, involving active participation from all members during the Hacktoberfest event. OpenStreetMap, a collaborative mapping platform, forms the foundation for this project, aiming to aggregate and enhance geospatial data to foster improved accessibility and navigation within communities.

Inovus Labs, a collective of passionate and dedicated individuals, has come together to contribute to this innovative effort. During Hacktoberfest, a month-long celebration of open-source contributions, team members pooled their expertise, time, and creativity to enrich the OpenStreetMap database.

Participants

All the students of Kristu Jyoti College of Management and Technology

8x8 LED Matrix



DESCRIPTION

An 8x8 LED matrix created by college students is a compact electronic display consisting of 64 individual Light Emitting Diodes (LEDs) arranged in an 8 by 8 grid pattern. The matrix serves as a visually dynamic platform for showcasing various patterns, text, or simple animations by selectively illuminating specific LEDs.

Components used:

- Arduino Nano
- LED
- PCB

Participant

1. Indhuchoodan R

Blind Stick



DESCRIPTION

A "Blind Stick" designed and developed by college students is a revolutionary assistive device aimed at enhancing the mobility and independence of individuals with visual impairments. This innovative tool incorporates cutting-edge technology and thoughtful design to address the specific challenges faced by those with limited or no vision.

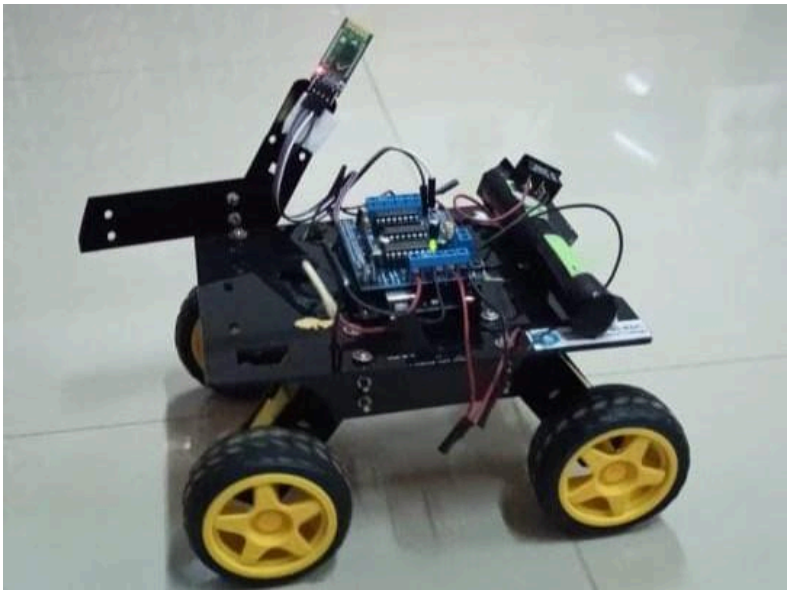
Components Used

- Arduino uno
- Ultrasonic Sensor
- buzzer
- stick

Participants

1. Abhishek V Gopal
2. Badhusha Shaji

Bluetooth RC Car



DESCRIPTION

Introducing our Bluetooth Remote Control Car designed by a group of talented college students with a passion for technology and public safety. This remarkable device is tailored specifically for fire rescue missions, revolutionizing the way emergency messages are delivered and received in high-intensity situations.

Components Used:

- Motorshield
- Arduino Uno
- Li-Po Battery
- Switch
- Car Body Kit
- Gear Motor

Participants

1. Abhishek V Gopal
2. Badhusha Shaji
3. Jeevan Joseph

RFID Based Smart Lock



DESCRIPTION

Here's an interesting tale behind the development of an RFID-based Smart Access Door (frankly speaking, an over-engineered one), that we've put together recently at Inovus Labs.

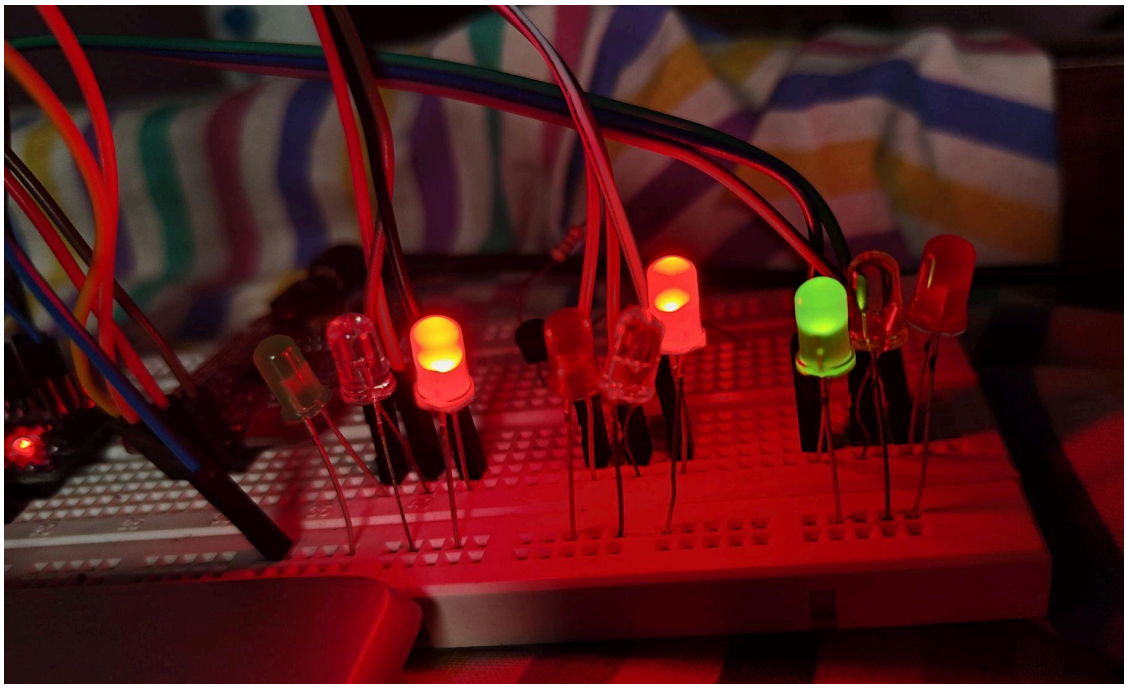
Components used:

- Hardware Requirements:
    - ESP32 Development Board
    - RFID reader
    - Magnetic switch
    - LED Indicators
    - Buzzers
    - 12v Solenoid Lock
    - Single Module Relay
    - JHD204A LCD Display
- Software Requirements
    - Firebase Realtime Database
    - Discord Webhooks

Participants

1. Abhishek V Gopal
2. Badhusha Shaji

Smart Traffic Lights using Voice Recognition Module



DESCRIPTION

The Smart Traffic Lights with Voice Recognition Module is an innovative and life-saving project developed by a group of dedicated college students. Designed with the primary goal of facilitating the swift and safe passage of ambulances through congested urban traffic, this intelligent traffic management system represents a breakthrough in emergency response technology.

Components used:

- LED Bulb
- Voice Recognition Module
- Arduino Nano
- Jumper Wire

Participants:

1. Abhishek V Gopal
2. Badhusha Shaji

E-waste Bin



DESCRIPTION

The E-waste bin is a pinstop solution for waste collection. The ultrasonic sensor HC-SR04 is used by the waste bin to identify items in front of it. The signals are subsequently transmitted to Arduino Uno. The Servo motor opens the flap on top of the trash can after the arduino recognizes the signal and sends a signal to it.

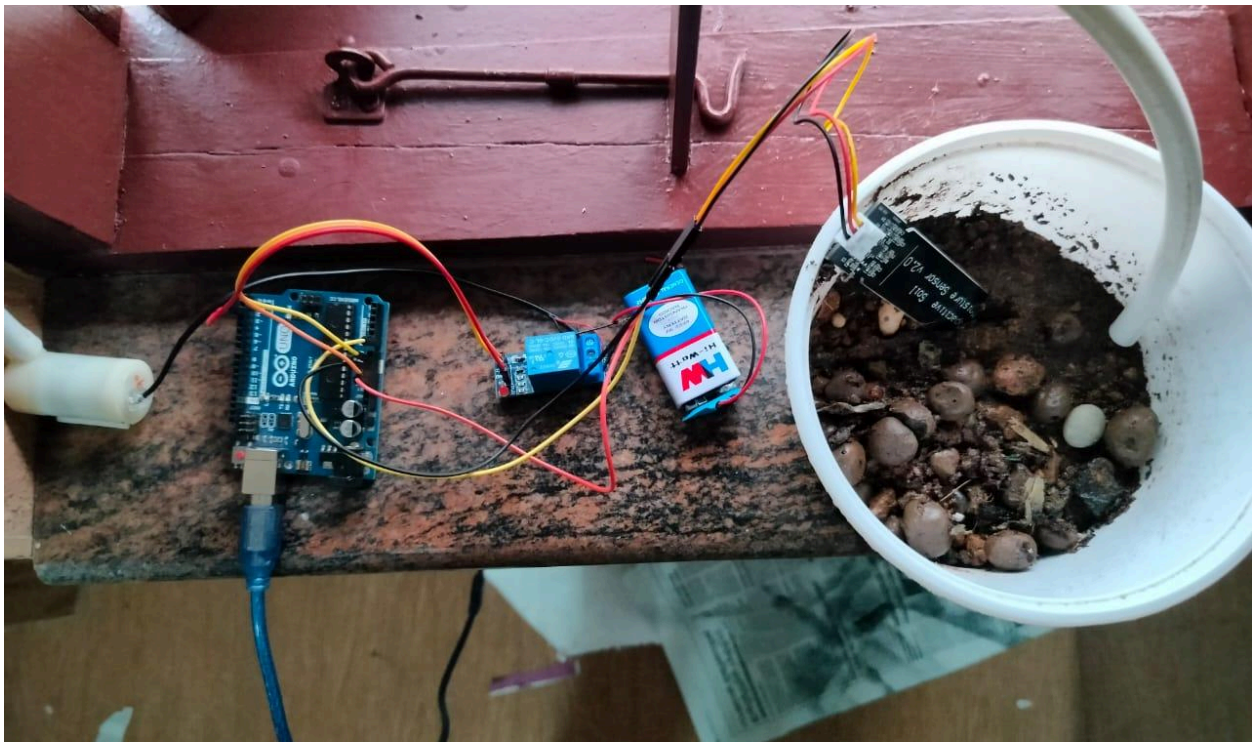
Components:

- Arduino Uno
- Ultrasonic Sensor
- Servo Motor
- Jumper Wires

Participants:

1. Abhishek V Gopal
2. Badhusha Shaji

Smart Garden using Capacitive Soil Moisture



DESCRIPTION

A smart garden using capacitive soil moisture technology is an innovative and efficient gardening system designed by a team of students. This is a better solution for gardening that integrates capacitive soil moisture sensors to optimize the watering process for plants.The project is aimed to reduce the water consumption by the individuals and to automate the process of watering.

Components:

- Capacitive Soil Moisture Sensor
- Arduino
- 5v Relay
- 12V DC water Pump
- Jumper Wires
- LED

Participants:

1. Nikhil T. Das
2. Indhuchoodan R

EduBulb



DESCRIPTION

The Education Bulb Workshop is a brilliant initiative crafted by a group of talented and innovative college students. This enlightening workshop is designed to empower learners of all ages with a unique and engaging educational experience.

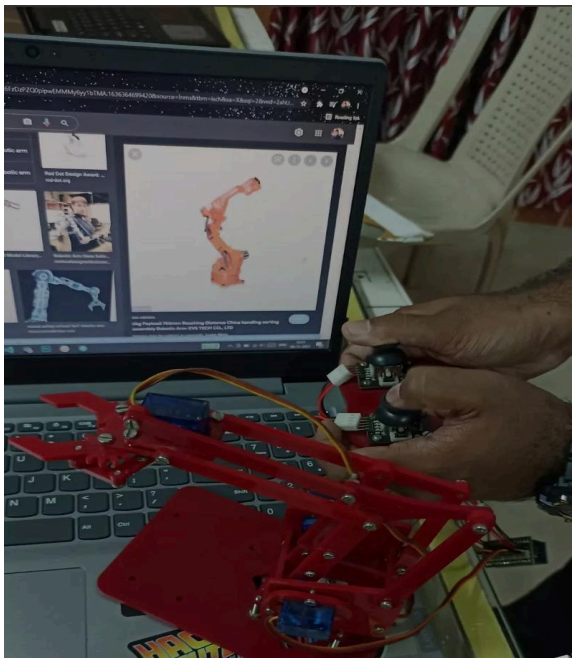
Components used:

- Tikki Fitting Machine
- Punching Machine
- Soldering Iron

Participants

1. Indhuchoodan R
2. Nikhil T Das

Robotic Arm



DESCRIPTION

A remarkable robotic arm, meticulously crafted by a team of ingenious college students, showcases an impressive fusion of engineering prowess and innovative design. This marvel of modern technology consists of multiple high-precision servos working in perfect harmony to execute intricate tasks with precision and finesse.

Components used:

- Arduino Nano
- Robotic Arm Kit
- Micro Servo

Participant

1. Arjun Krishna

Breath Analyzer



DESCRIPTION

The Breath Analyser created by college students is a cutting-edge, portable device designed to accurately measure and analyze the composition of an individual's breath. The Breath Analyser can swiftly and precisely detect the presence of alcohol in a person's breath, making it a crucial tool for law enforcement agencies to combat drunk driving and ensure road safety.

Components used:

- MQ-3 Sensor
- Arduino Nano
- LED
- buzzer

Participant

1. Jeevan Joseph

LPG Detector

DESCRIPTION

A college student-created LPG (liquefied petroleum gas) detector is an innovative device designed to enhance safety in domestic and commercial environments by detecting the presence of potentially hazardous gas leaks. This compact and user-friendly device is the result of collaborative efforts and cutting-edge engineering from a group of talented and ambitious college students.

Components used:

- Arduino Uno
- MQ-6 Sensor
- LED
- Buzzer

Participants

1. Nikhil T Das
2. Jeevan Joseph